Katie Ellman-Aspnes

South Bend, IN

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Education

University of Notre Dame

Ph.D. in Mathematics — August 2021 – May 2027 (expected)
Advisor: Nick Ramsey

M.S. in Mathematics — Dec 2022 (degree conferred Jan 2027)
Advisor: Nick Ramsey

Vassar College

Bachelor of Arts in Mathematics & Statistics and Computer Science — August 2016 – May 2020
Advisors: Charles Steinhorn (mathematics) and Marc Smith (computer science)

Research

Research Interests: Model theory, classification theory, NSOP_1 theories, independence and dimension.

Preprint

- Approximation by simple structures, 2026. [arXiv]

Invited Talks

Jerusalem Logic Seminar

Approximation by simple structures — Dec 2, 2026 (upcoming)

AMS Fall Central Sectional Meeting: Model Theory Special Session

Approximation by simple structures — Oct 24, 2026

UIUC Logic Seminar

Approximation by simple structures — Oct 6, 2026

Wesleyan/UCONN Logic Seminar

Approximable Structures — Apr 8, 2025

Vassar Mathematics Colloquium

Classification Theory and the Model-Theoretic Map of the Universe — Apr 3, 2025

Other Talks and Presentations

ASL North American Meeting

Approximating NSOP_1 Structures — Jun 20, 2026

26th Graduate Student Conference in Logic

Approximating NSOP_1 Structures — Apr 25, 2026

Notre Dame Model Theory Seminar

Approximation by simple structures — Sep 22, 2026
Dimension in approximated structures — Oct 28, 2025
Approximating models of NSOP_1 theories via substructures with simple theories — Sep 24, 2024
The Independence Theorem in simple theories — Jan 31, 2023
Elimination of $\exists^\infty$ in pregeometric fields — Aug 30, 2022

Notre Dame Graduate Student Seminar

Classification Theory and the Map of the Universe — Feb 20, 2026

Notre Dame Graduate Student Logic Seminar

Model-theoretic properties of Conant independence — Dec 7, 2023
Free amalgamation and NSOP_4 — Oct 26, 2023
Basic definitions in mathematical logic — Sep 15, 2022

Notre Dame Mathematics Oral Ph.D. Candidacy Exam

The Independence Theorem in simple theories — Dec 7, 2022

Teaching Experience

University of Notre Dame

Instructor of Record
- MATH 10550, Principles for Calculus ..... Fall 2025
- MATH 10560, Calculus II Online (course chair) ..... Summer 2025
- MATH 10560, Calculus II Online ..... Summer 2024

Teaching Assistant
- MATH 10550, Principles for Calculus (Head TA) ..... Fall 2024
- MATH 10560, Calculus II (Head TA) ..... Fall 2023
- MATH 10560, Calculus II ..... Spring 2023
- MATH 20550, Calculus III ..... Fall 2022

Guest Lecturer
- MATH 10130, Beginning Logic ..... Mar 16, 2026; Mar 20, 2026
- MATH 30610, Introduction to Financial Mathematics ..... Feb 3, 2025; Feb 7 2025
- MATH 10130, Beginning Logic ..... Sep 20, 2023; Sep 22, 2023

Vassar College

Teaching Assistant
- CMPU 224, Computer Organization ..... Fall 2018–Spring 2020
- CMPU 101, Problem Solving and Abstraction ..... Fall 2017–Spring 2018

Honors & Awards

University of Notre Dame

Kaneb Center for Teaching Excellence
- Advanced Teaching Scholar Workshop Program ..... September 2026
- Striving for Excellence in Teaching ..... December 2025

Notre Dame Learning
- Effective Online Pedagogy Digital Badge ..... May 2024

Vassar College

Mary Evelyn Wells & Gertrude Smith Prize May 2020

Institutional Honors
- Departmental Honors, Mathematics ..... May 2020
- Departmental Honors, Computer Science ..... May 2020
- General Honors ..... May 2020

Service

University of Notre Dame

25th Graduate Student Conference in Logic
- Co-organizer ..... April 2025

Notre Dame Mathematics Graduate Student Association
- President 2023–2025
- First-Year Representative 2021–2022

GradientND
- Vice President 2024–2025
- Secretary 2023–2024